

Neural geometry of an RL agent’s beliefs and behavior: methods

Abstract

This document specifies, step by step, the analysis pipeline we use to expose the *belief*, *decision* and *behavior* structure inside a recurrent RL agent. The pipeline constructs three objects — a **belief space** (distributions over the latent task variable), a **behavior space** (distributions over future outcomes), and an **activation manifold** (the geometry of the hidden states that carry both) — and then intervenes on them in both directions: editing beliefs to change behavior, and forcing behavior to change beliefs. Every technique operates on plain arrays (hidden states, labels, progress, distributions) and is therefore portable to any navigation-and-decision environment; the only environment-specific component is the data-collection layer.

1 Setting and notation

An agent with a recurrent policy interacts with a POMDP. At time t it receives an observation o_t , updates its hidden state

$$\mathbf{h}_t = F(\mathbf{h}_{t-1}, o_t) \in \mathbb{R}^d, \quad (1)$$

(here F is a GRU, $d=256$) and samples an action $a_t \sim \pi(\cdot | \mathbf{h}_t)$. The environment contains a *latent task variable* c (in MemoryEnv: the cue identity, $c \in \mathcal{C}$, which factorizes into direction and color) that is observable only during a brief window, and is *consumed* by decisions much later. We also fix a scalar **progress parameter** x_t (in MemoryEnv: the agent’s corridor column; in a noisier environment any monotone task-progress proxy works, e.g. distance-to-goal or a learned progress index).

Portability requirements. To run the pipeline on another environment (e.g. the `bridge_tunnel` navigation task) one must provide: (i) rollouts with hidden states $\mathbf{h}_t$; (ii) a label for the latent variable of interest c (what the agent must remember); (iii) a progress parameter x_t ; (iv) a definition of terminal *outcomes* z (which decisions were taken). Nothing else in the pipeline changes.

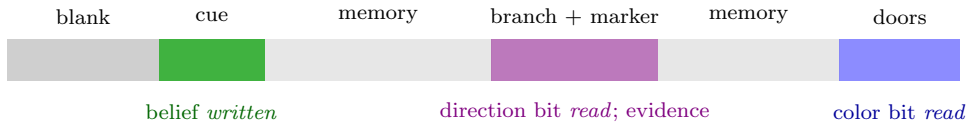


Figure 1: One episode as a memory pipeline: a two-bit belief is written once, carried, and its bits are spent at two different decision points. The mid-corridor marker door provides branch-identity *evidence* that the agent experiences en route.

2 Data collection

Traces. We roll the *stochastic* policy (actions sampled from the softmax) for n episodes per training cue and store, per live step, the tuple $(\mathbf{h}_t, x_t, c, t, \text{cue-seen}_t)$, where cue-seen_t flags whether the cue has entered the agent’s egocentric view at any $t' \leq t$. Sampling (rather than greedy) matters: it broadens the state distribution around each trajectory, which manifold fitting and probing need; greedy rollouts collapse onto a handful of paths.

Alignment convention. $\mathbf{h}_t$ is the hidden state *after* processing o_t ; the pair $(\mathbf{h}_t, s_{t+1})$ — belief after seeing o_t , environment state after acting — is the correct point to resume simulation from (used by the Monte-Carlo behavior estimates below).

3 Belief space

Probe. A multinomial logistic probe estimates the agent’s posterior over the latent variable from the hidden state:

$$P(c \mid \mathbf{h}) = \text{softmax}(\mathbf{W}\mathbf{h} + \mathbf{b})_c, \quad (2)$$

trained with cross-entropy + ℓ_2 on trained-cue states. Crucially the class set is $\mathcal{C} \cup \{\emptyset\}$: a **blank class** of pre-cue states ($\text{cue-seen} = \text{false}$). Without it the probe must extrapolate on pre-cue states that lie off its training manifold, and logistic extrapolation is confident by construction — producing the artifact that “the belief is already decided before the cue”. With the blank class, $P(\emptyset \mid \mathbf{h}) \approx 1$ before the cue and control transfers to the cue classes exactly at cue onset (§10).

The space. Each state now maps to a point on the probability simplex $\Delta^{|\mathcal{C}|}$. We call the image of the data under $\mathbf{h} \mapsto P(\cdot \mid \mathbf{h})$ the *belief space*.

4 Behavior space

The behavior of a state is not its next action but the *distribution over futures* it induces. For a stratified subset of states (a few per progress column per cue) we resume the environment from $(\mathbf{h}_t, s_{t+1})$ and roll the stochastic policy K times ($K=24$), recording the terminal outcome $z \in \{\text{up, down}\} \times \{\text{green, blue}\}$:

$$\hat{P}(z \mid \mathbf{h}_t) = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[z^{(k)} = z]. \quad (3)$$

This **Monte-Carlo outcome posterior** lives on Δ^4 and defines the *behavior space*. Early states have high-entropy behavior only through policy stochasticity; after each decision is spent, outcome mass collapses — the geometry of the behavior space literally displays decision-making.

5 inPCA: embedding spaces of distributions

Belief and behavior points are probability vectors, so Euclidean PCA is the wrong geometry. We embed them with *intensive PCA* (inPCA; Quinn et al., PNAS 2019), the technique used in Goodfire’s “Shape of Beliefs”. Given points $\{p^{(i)}\}$, form the pairwise Bhattacharyya divergence

$$D_{ij} = -\log \sum_k \sqrt{p_k^{(i)} p_k^{(j)}}, \quad (4)$$

double-center it, $W = -\frac{1}{2}JDJ$ with $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, and eigendecompose $W = V\Lambda V^\top$. Coordinates are $V\sqrt{|\Lambda|}$, keeping eigenvalues *signed*: inPCA is an isometric embedding into Minkowski

space (negative- λ directions are “time-like”); in practice the leading components are positive and the plot reads like PCA but with distances that are faithful to probability space. Where only a metric (not a visualization) is needed we use the closed-form **Hellinger map** $p \mapsto \sqrt{p}$, under which the simplex embeds almost-Euclideanly ($d_H^2(p, q) = \sum_k (\sqrt{p_k} - \sqrt{q_k})^2$).

Why not UMAP/t-SNE? Both are non-parametric: they cannot project *new* states (e.g. steered, off-distribution ones) faithfully, they distort distances by design, and their layout is fit-dependent — all fatal for before/after intervention comparisons. Every projection we use (probes, PCA, inPCA, Hellinger) is parametric and out-of-sample-safe.

6 The activation manifold: a tree of fibers

For each cue c and progress bin x compute the centroid $\mu_c(x) = \mathbb{E}[\mathbf{h} \mid c, x, \text{cue seen}]$, and for pre-cue states the cue-independent centroid $\mu_\emptyset(x)$. Connecting centroids along x yields polyline *fibers*, one per cue, plus a shared pre-cue *trunk*: the activation manifold of this task family is a **tree** (Fig. 2). Fiber arc-length $s_c(x) = \sum_{x' \leq x} \|\mu_c(x'+1) - \mu_c(x')\|$ gives geodesic distances *on* the manifold:

$$d_{\text{tree}}((c, x), (c', x')) = \begin{cases} |s_c(x) - s_{c'}(x')| & c = c' \\ s_c(x) + s_{c'}(x') & c \neq c' \quad (\text{path through the split}), \end{cases} \quad (5)$$

with arcs measured from the split point (cue-room exit). Splines can replace polylines when bins are dense; for noisier environments the same construction applies with soft binning of x .

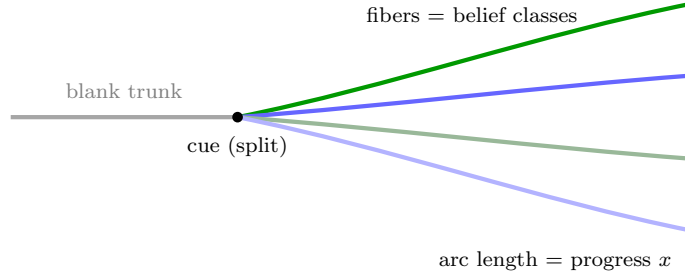


Figure 2: The activation manifold as a trunk-and-fibers tree. “Which fiber” is the belief; “where along the fiber” is task progress. Geodesics between fibers pass through the split.

Isometry test. Following Goodfire’s manifold-steering analysis we correlate tree geodesics with distribution distances: for all node pairs, plot d_{tree} against the Hellinger distance of the nodes’ mean *belief* distributions and of their mean *behavior* distributions. The expected signature: belief distance is a step function of the tree (zero within a fiber, saturated across fibers — belief is *which* fiber), while behavior distance grows along fibers as decisions are consumed. This is the geometric statement of the belief/behavior distinction.

7 Linear field probes: does the memory code rotate?

A single probe assumes one global belief axis. A *linear field probe* (LFP; “Shape of Beliefs”) is a family of probes indexed by progress: for every bin x fit a multinomial probe W_x on the states *in that bin only*. Three diagnostics:

- **Gram matrix** $K_{xx'} = \cos(\text{vec}(W_x), \text{vec}(W_{x'}))$: near-constant $\Rightarrow$ stationary code; banded/decaying $\Rightarrow$ the code *rotates* with position;

- **transfer matrix** $T_{x \rightarrow x'}$: accuracy of the bin- x probe evaluated at bin x' ;
- **rotation angle** $\theta = \arccos K_{x_{\min} x_{\max}}$ between the first and last probes.

A rotating code has a direct causal consequence: *any single global steering axis is the wrong axis somewhere along the corridor* — the explanation for interventions that decode perfectly yet fail causally.

Additivity. The centroid grid also admits a factorization test: fit $\mu_c(x) \approx m + a_c + b_x$ by least squares and report R^2 . High R^2 means memory content and progress are encoded in independent (additive) subspaces; the residual quantifies their interaction.

8 Interventions

8.1 Belief $\rightarrow$ behavior: transient swaps, global vs field-aware

A **global swap** implants cue t into an episode of cue s by clamping the hidden state along the class-mean axis during a window $\mathcal{W}$ (cue-room exit until the branch decision; released afterwards):

$$\mathbf{h} \leftarrow \mathbf{h} + (\rho - \mathbf{h}^\top \hat{\mathbf{u}}) \hat{\mathbf{u}}, \quad \hat{\mathbf{u}} = \frac{\mu_t - \mu_s}{\|\mu_t - \mu_s\|}, \quad \rho = \mu_t^\top \hat{\mathbf{u}}. \quad (6)$$

The **field-aware swap** replaces the single axis with a *position-local* field: per-column axes $\hat{\mathbf{u}}(x) \propto \mu_t(x) - \mu_s(x)$ and targets $\rho(x) = \mu_t(x)^\top \hat{\mathbf{u}}(x)$, spline-smoothed across columns (coordinate-wise), applied with the agent’s current column x_t . If the code rotates (§7), only the field version tracks the manifold.

Path energy. Interventions are scored not only by success (“behaves as target”: target branch *and* target door) but by *naturalness*: the mean Bhattacharyya distance of the probe posterior to the target vertex during the window, $E = \frac{1}{|\mathcal{W}|} \sum_{t \in \mathcal{W}} d_{\text{BC}}(P(\cdot | \mathbf{h}_t), \delta_t)$ — Goodfire’s manifold-energy criterion, discretized. Low energy means the edit moved the state *along* the belief manifold rather than through a void.

8.2 Behavior $\rightarrow$ belief: forcing through evidence

The converse intervention never touches $\mathbf{h}$: pure *action replacement* drives the agent through the direction-*wrong* corridor, including opening its marker door — an observation that never co-occurs with the true cue in training — then releases control. Readouts: the probe posterior aligned at the marker-opening event, the final door choice, and the **implied evidence weight**

$$\text{LLR} = \log \frac{P(\text{destination cue} | \mathbf{h}_{\text{pre-door}})}{P(\text{true cue} | \mathbf{h}_{\text{pre-door}})}, \quad (7)$$

measured at the last step before the final doors enter the agent’s view (so the readout reflects memory, not fresh door perception). A symmetric Bayesian observer holding contradictory cue-vs-corridor evidence of equal reliability sits at $\text{LLR} = 0$; $\text{LLR} > 0$ quantifies recency dominance. Whether the belief *collapses to another attractor* (entangled training set) or *revises one factor only* (factorized training set) is the central comparative result.

9 Attractor census

Beliefs persist because they are attractors of the closed-loop dynamics. Fix a neutral “maintenance” observation o^* (mid-corridor, nothing salient in view) and study the step map $\Phi(\mathbf{h}) =$

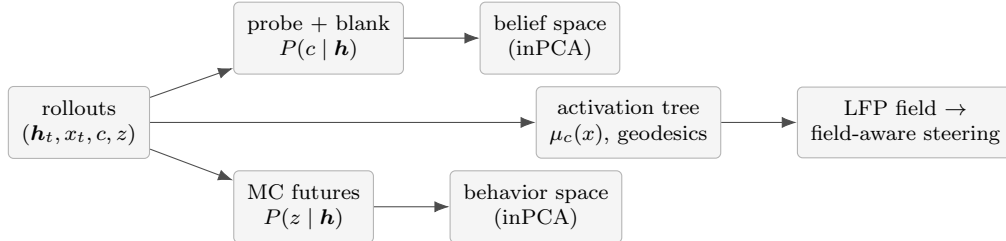
$F(\mathbf{h}, o^*)$. Fixed points are found by minimizing $\|\Phi(\mathbf{h}) - \mathbf{h}\|^2$ with Adam from a few hundred data-adjacent initializations, merging duplicates; stability follows from the spectral radius of the Jacobian $\partial\Phi/\partial\mathbf{h}$ at each point (<1 : attractor). Each fixed point is identified with its nearest class mean. The census (how many attractors, their stability margins, where the saddles sit) explains asymmetries in steerability: a “default” cue is a deeper attractor, and doses that fail to cross its basin boundary are erased by the flow.

Flow-field panels. For visualization we take a 2-D affine slice of state space through the mean hidden state, spanned by orthonormalized probe axes; each grid point of the slice is a genuine state $\mathbf{h}(u, v)$, colored by the probe’s decision regions and overlaid with the *in-plane* component of $\Phi(\mathbf{h}) - \mathbf{h}$ as arrows. (Caveat to keep in the figure caption: motion out of the slice is invisible; the slice shows a section of a d -dimensional flow.)

10 The pre-cue bias: diagnosis and fix

Symptom. Probes report $P(\text{one cue}) \approx 1$ *before* the cue is visible. **Diagnosis.** Two stacked causes: (i) a probe trained only on post-cue states is evaluated off its training manifold; logistic regression partitions all of $\mathbb{R}^d$, so off-manifold points get confident, arbitrary labels; (ii) genuinely, the pre-cue trunk lies closer to one fiber — the network’s *default basin* (an honest, measurable fact about its dynamics, not a bug). **Fix (no retraining).** Add the blank class (§3) and report the distance-to-manifold $d(\mathbf{h}) = \min_{\text{nodes}} \|\mathbf{h} - \mu\|$, normalized by the on-manifold median, alongside every belief readout: pre-cue states then read as “blank” and any steered state that leaves the manifold is flagged rather than silently misread. A retraining fix (auxiliary calibrated-belief loss) would inject representation supervision and break the emergent-concepts design — we deliberately do not use it.

11 Summary of the pipeline



The same seven analyses (belief space, behavior space, activation tree, isometry, LFP/rotation, two-way steering with path energy, attractor census) apply verbatim to any environment providing the four ingredients of §1 — for **bridge_tunnel/cogniland** the latent variable is the committed skill and map category, progress is distance-to-target, and outcomes are (skill used, target reached).